



OPEN Imbalanced feature generation based on bootstrap power spectral curve for estimating respiratory rate

Soojeong Lee¹, Gyanendra Prasad Joshi² & Gangseong Lee³✉

Rapid respiratory rate (RR) changes in older adults may indicate serious illness. Therefore, accurately estimating RR for cardiorespiratory fitness is essential. However, machine learning algorithm-related errors are unsuitable for medical decision-making processes because some data have a much larger sample size in the training set than in other sets. This difference in size refers to data imbalance. Therefore, we introduce a novel methodology combining bootstrap-based imbalanced feature generation (BIFG) with the Gaussian process for estimating RR and uncertainty, thereby addressing data imbalance. The sample difference between normal breathing (12–20 bpm), dyspnea (≥ 20 bpm), and hypopnea (< 8 bpm) indicates significant data imbalance, which can affect the learning of the machine learning algorithm. Thus, the normal breathing part with much data is well-trained. The dyspnea and hypopnea parts with relatively little data are not well-trained, and this data imbalance causes significant errors concerning the reference variables in the actual dyspnea and hypopnea data parts. Hence, we use the parametric bootstrap model generated by artificial feature curves to estimate RR and solve this problem. As a result, the non-parametric bootstrap approach drastically increased the number of artificial feature curves. The generated artificial feature curves are selectively utilized for the highly imbalanced parts. Therefore, BIFG can be efficiently trained to predict the complex nonlinear relationships between the feature vectors obtained from the photoplethysmography signals and the reference RR. The proposed methodology exhibits more accurate predictive performance and uncertainty. The mean absolute errors are 0.89 and 1.44 beats per minute for RR using the proposed BIFG based on the two data sets.

Keywords Respiratory rate estimation, Confidence interval, Gaussian processing regression, Power spectral curves, Bootstrap, Photoplethysmography

Respiratory rate (RR) monitoring has emerged as an effective health management indicator for individuals with respiratory diseases, particularly for those who live alone¹. In older individuals, rapid changes in RR suggest serious illness^{2,3}. Tachypnea is a fast, shallow RR in adults, usually defined as more than 20 breaths per minute (bpm) while resting. It can indicate an underlying health problem that requires medical attention, especially if other symptoms like chest pain, dizziness, or fatigue accompany it⁴. Hence, accurately estimating RR for cardiorespiratory health is vital. At present, capnography is the standard method for measuring RR. However, this method is expensive and requires strict management⁵.

Body movement increases the RR and rapidly changes the body state³; therefore, it is commonly used to detect respiration using non-contact sensors. However, currently, available non-contact RR measurement systems have limitations in representing this variability because they employ a single measurement value. Applying a pulse oximeter with oxygen saturation (SpO₂) and heart rate to obtain an accurate RR is both patient-friendly and economical⁶. Biomedical and engineering researchers have used photoplethysmography (PPG) signal-based wearable devices to predict SpO₂. Furthermore, PPG and electrocardiography (ECG) sensor-based continuous signal measurement approaches have been employed to estimate RR^{7,8}.

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Over the past decade, machine learning (ML) has been widely applied in biomedical engineering to estimate objective values^{1,9,10}. Lee et al.¹¹ presented an ensemble gradient boosting algorithm (EGBA) with PPG signals for estimating RR. Furthermore, multiphase feature-based EGBA has improved RR estimation¹¹. However, if multiple feature extraction methods are utilized, inappropriate feature vectors may be selected based on the feature extraction method, diminishing the reliability of RR estimation¹¹. Thus, a practical feature extraction method can be incorporated into an ML algorithm in the automatic feature extraction process.

Additionally, computer resource loss may occur when multiple transformation methods are utilized in the feature extraction⁹. Liu et al.¹² introduced generative boosting via long short-term memory (LSTM) for estimating RR. Subsequently, Kumar et al.¹³ applied LSTM to estimate RR using PPG and ECG signals. LSTM is primarily used in time domain and image data processing. However, LSTM has the following limitations: a complex structure, use of several parameters, severe fluctuations in the predictive performance based on the data used, and excessive consumption of computer resources⁹. Support vector machine (SVM) algorithm assumes a kernel that determines the mapping between the feature and target vectors. Then, a training dataset is utilized to estimate the unknown parameters of the kernel function. Singh et al.¹⁴ used to integrate the SVM as a simple structure based on the radial basis function kernel for automatic classification of asthma and non-asthma. Thus, these algorithms are different from parametric algorithms.

Gaussian process regression (GPR) is another non-parametric approach¹⁵ that does not require parameters for input-to-output mapping. GPR uses a series of values, rather than a single value, as an estimate and provides direct confidence intervals (CIs) to measure the uncertainty of RR estimations. This uncertainty cannot be obtained directly from neural networks (NN)¹⁶ or LSTM¹³. Hence, the CIs obtained through the GPR model estimate the uncertainty of continuous RR monitoring. However, the CIs obtained from the GPR model are calculated based on the distribution of RR estimates, which provides a relatively wide CI. Although the probability that the RR estimate is included in relatively wide CIs increases, it has the disadvantage of reducing the reliability of the RR monitoring system. In addition, if the CI is too narrow, even a tiny change in the RR estimate can easily lead to an out-of-CI, which limits its role as an RR monitoring system. Securing an appropriate range of CIs is necessary to overcome these shortcomings.

The reliability and accuracy of ML algorithms can be increased by extracting and determining the features vital for the target variable¹⁰. Thus, several researchers have developed feature extraction algorithms^{10,17–20}. Recently, Lee et al.⁹ developed a novel method for estimating RR using GPR with hybrid feature extraction using PPG and ECG signals. Power spectrum and multiphase algorithm-based features were utilized to compensate for the limited data available. Lee et al.^{10,21} introduced an accurate Gaussian process-based integrated automatic feature extraction technique for RR and uncertainty estimation.

ML algorithms trained using limited data, including the NN and LSTM models, are generally sensitive to the training sequence and initial parameter values, resulting in unstable performance²². Therefore, like most ML algorithms, collecting reasonably sized data is vital for training GPR models. However, in the biomedical field, data collection encompasses several challenges and practical limitations. First, patient data collection requires significant time and effort. Second, it is often difficult to obtain many patients with a particular disease owing to its low prevalence²³. Therefore, research on the effects of limited sample size is attracting attention.

Furthermore, ML errors associated with using imbalanced data have garnered less attention, which has primarily been covered in the literature in the field of ML^{24,25}. Data imbalance can drastically affect the training of traditional ML algorithms²⁶. It involves both convergence during the training step and generalization of an algorithm on the test set²⁴. Two representative models augment imbalanced data sets in classification models using ML algorithms. Under sampling methods, most class samples are randomly removed to balance the class distribution. However, eliminating the majority of class samples can result in information loss and deteriorate the classifier's performance. Over-sampling methods randomly replicate samples of the minority class to balance the class distribution²⁷.

Douzas et al.²⁸ proposed a simple and efficient over-sampling methodology that combines k-means clustering and synthetic minority oversampling techniques. Sun et al.²⁹ presented an over-sampling method that effectively addresses the challenge of negative over-sampling results to generate new synthetic samples in the minority class region. However, over-sampling also has drawbacks, such as the possibility of overfitting, introducing noise into the dataset, increasing computational complexity, and limiting information gain. These shortcomings may affect the generalization ability of the ML and its performance in the test environment.

Until recently, data imbalance studies have mainly focused on data classification. However, studies have yet to be found that address the data imbalance problem to improve the accuracy of the RR estimation and estimate CIs, which are indicators of uncertainty. Data imbalance still existed in the feature dataset. For example, the sample difference between normal breathing (12–20 bpm), dyspnea (≥ 20 bpm), and hypopnea (< 8 bpm) indicates significant data imbalance³⁰, which is a serious problem when using the GPR algorithm, because data imbalance can affect the learning of the GPR algorithm. In more detail, the normal breathing samples with much data are well-trained. In contrast, the dyspnea and hypopnea samples with relatively little data are not well-trained, and this data imbalance causes significant errors concerning the reference variables in the actual dyspnea and hypopnea data parts.

Based on this motivation, we introduce a novel method, bootstrap-based³¹ imbalanced feature generation (BIFG), to estimate RR and uncertainty using the GPR algorithm to solve the problems mentioned above in this study. The BIFG method, which is a unique approach to handling imbalanced datasets, involves creating multiple resampled datasets. This flexible, non-parametric approach³¹ estimates variability across different subsets of the data, helping to produce more stable and reliable results, especially in the face of nonlinear relationships and data with high variance³². In our research, the BIFG method generates continuous power spectral artificial curves using the imbalanced data signal samples, i.e., dyspnea and hypopnea signals. The generated power spectral artificial curves are then fused with the power spectral curves extracted from the original dyspnea and

hypopnea signals and used as training data. As a result, the proposed BIFG method dramatically increases the number of artificial power spectral curves as features. These artificial power spectral curves are selectively used for severely imbalanced feature sets. Increasing the number of artificial power spectral curves for dyspnea and hypopnea samples can make the dataset more balanced for the GPR algorithm. Hence, GPR can better learn the characteristics and patterns of dyspnea and hypopnea.

While ML and deep learning (DL) methods have been widely used for RR estimation, our work presents a unique approach that addresses a critical gap—handling imbalanced datasets effectively through BIFG integrated with GPR. Unlike conventional ML and DL models, which struggle with class imbalance, our method improves model generalization and uncertainty estimation without requiring large amounts of labeled training data. Additionally, BIFG enhances feature representation by generating artificial power spectral curves, leading to a more balanced dataset and improved model convergence. Our approach ensures that underrepresented classes, such as dyspnea and hypopnea, which are often overlooked in standard GPR applications due to their low occurrence in the dataset, contribute more effectively to the learning process, addressing biases in standard GPR applications. By directly tackling dataset imbalance and enhancing GPR's predictive stability, our method significantly advances RR estimation, making it more reliable for real-world clinical applications.

Consequently, GPR can be efficiently trained to predict the complex nonlinear relationship between PPG signals' fused power spectral curves and the reference RR. This is the first study using GPR with imbalanced feature curves to estimate RR. The concept of the proposed algorithm is that RR and uncertainty are directly estimated using the GPR combined with the BIFG algorithm. Compared with¹⁰, this study makes the following contributions: Figure 1 shows a block diagram of the BIFG algorithm for estimating RR.

- A new method for estimating RR and uncertainty was proposed using the BIFG algorithm based on GPR algorithm.
- The non-parametric bootstrap approach was utilized to generate artificial feature curves for estimating RR to address severely imbalanced features.
- The proposed methodology is helpful in clinical settings or diagnostic applications for estimating RR and uncertainty by providing a CI and low estimated mean absolute error, possibly suggesting more accurate predictive performance and uncertainty.

Data set

Collection of PPG signals

We used two public data sets to evaluate the proposed algorithm. The first dataset used in this study was extracted from the MIMIC-II resource³³ of the Beth Israel Deaconess Medical Center (BIDMC)³⁴. It comprises ECG, PPG, and impedance pneumography (IP) respiratory signals acquired from patients aged 19–90 and admitted to an intensive care unit, with 53 recordings at 125 Hz over 8 min. The patients in the dataset were randomly selected

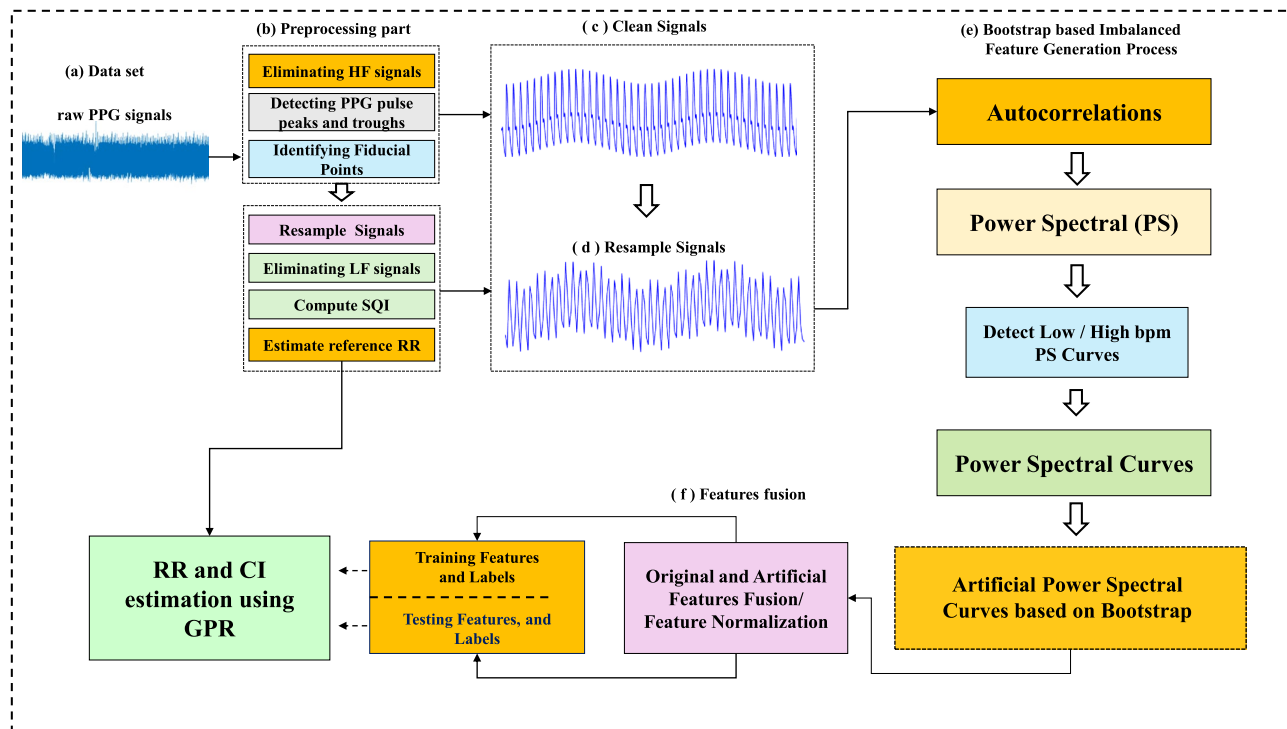


Fig. 1. The proposed BIFG methodology with GPR for RR and uncertainty estimations, where (a) denotes a raw PPG signal, (b) is a preprocessing part, (c) denotes a clean signal, (d) resample signal, (e) bootstrap-based imbalanced feature generation process, and (f) feature fusion.

from a cohort of individuals hospitalized at the BIDMC in Boston, USA. Reference RR values were derived using two sets of annotations for individual respirations in the IP signal. As shown in Fig. 2a, the distribution of RR reference differences between normal breathing (12–20 bpm), dyspnea (≥ 20 bpm) and hypopnea (< 8 bpm) indicated significant data imbalance³⁰. We presented an extended distribution of BIFG-based reference RRs on the training dataset, such as the red dash lines in Fig. 2b. We also denoted a distribution of RR reference on a testing dataset in Fig. 2c. We also presented an expanded distribution of BIFG reference RR on the testing dataset, such as the red dash lines in Fig. 2d, based on the BIDMC data set.

The second RRSYNTH data set³⁴ consists of 192 wave signals, each of which consists of a record length of 210 seconds at a sampling frequency of 500 Hz. The RRSYNTH data set is an artificial synthetic PPG signal modulated and generated in three ways to reflect the characteristics of various cardiovascular and respiratory diseases. First, baseline modulation (BW) represents the baseline change of the PPG signal well. Second, amplitude modulation (AM) represents the amplitude change of the signal. Third, the interval of one cycle of the PPG signal waveform varies depending on the frequency modulation (FM). Therefore, a total of 192 records of three modulation signals, BW (64 records), FM (64 records), and AM (64 records), were used to test various respiratory diseases. The respiratory signal's physiological mechanisms are described in³⁴. PPG BW modulation is known to be caused by changes in blood volume in the tissue. Blood volume changes are caused by changes in thoracic pressure delivered through the arteries and vasoconstriction of the arteries that carry blood to the veins during inspiration. AM indicates decreased stroke volume during inspiration due to changes in intrathoracic pressure. FM also indicates a spontaneous increase in HR and a decrease in expiration during inspiration, a frequent signal phenomenon when it is a symptom of respiratory arrhythmia (RSA)³⁵. Additionally, leveraging public datasets ensures the reproducibility and comparability of our results with future studies.

Preprocessing steps

In biomedical and related research fields, PPG has become a popular tool for estimating RR, heart rate, and SpO₂ and predicting blood pressure. Notably, it has revolutionized our understanding and the monitoring of various physiological parameters in the body. PPG signals were collected using the MIMIC-II and RRSYNTH data sets mentioned in the previous subsection. Next, high-frequency signals were removed using a Kaiser window with a cutoff frequency of 35 Hz and signal bandwidth of 3 dB. A merge segmentation technique was used to segment the PPG signal into multiple pulses, as described in³⁶. The reference points were obtained from the highest and lowest points of the PPG signal (Fig. 1b). Similarly, the lowest points were obtained. Next, linear interpolation

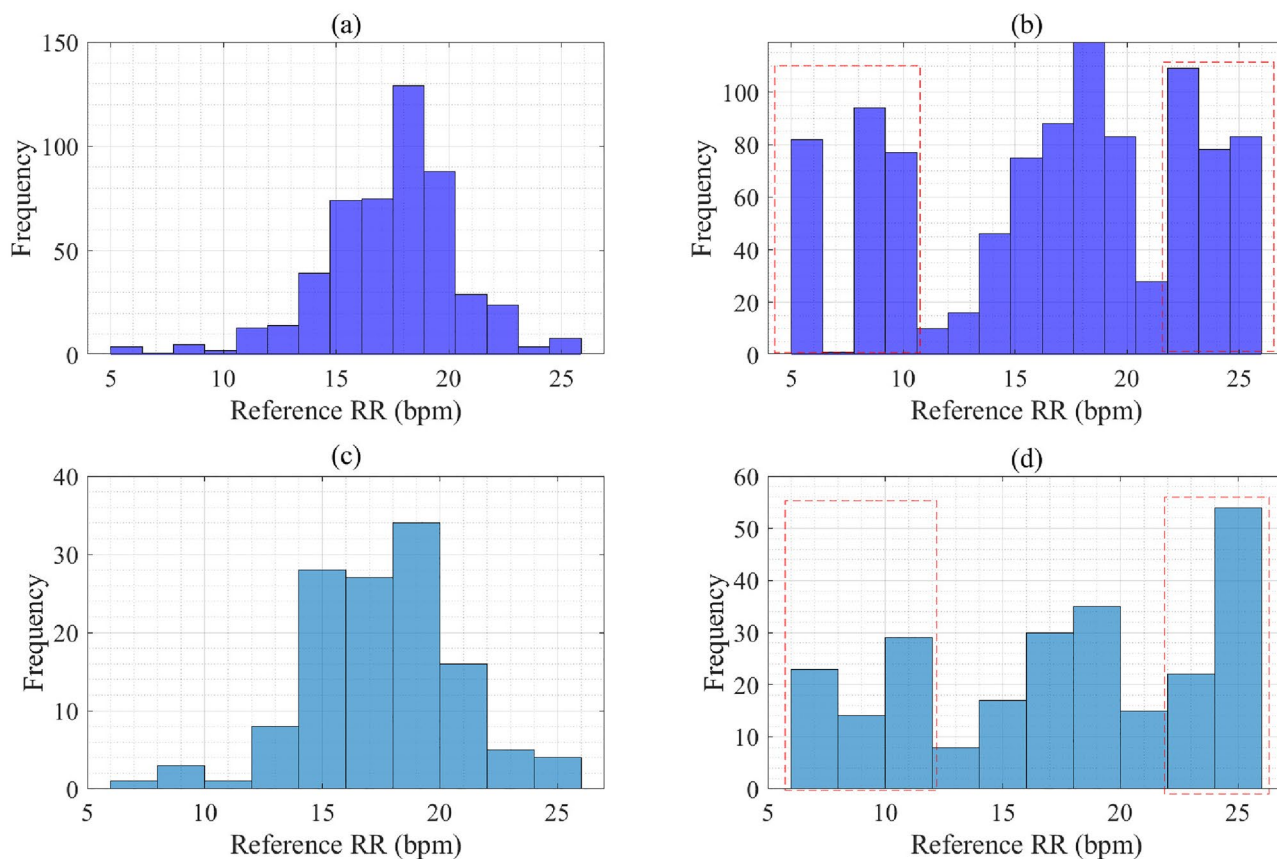


Fig. 2. The distribution of reference RR values. Panel (a) denotes a distribution of reference RR for training; panel (b) is an extended distribution of reference RR using BIFG for training; Panel (c) denotes a distribution of reference RR for testing; panel (d) is an extended distribution of reference RR using BIFG for testing.

was used to resample the PPG signal at 5 Hz, and low-frequency PPG signals were removed using a low-pass filter (Kaiser window function) with a cutoff frequency of 0.0665 Hz and bandwidth of 3 dB (Fig. 1b). Finally, the PPG signals yielded a resampled waveform. The resampled waveform dataset was obtained from individual records or participants. The mean seconds between continuous breathing in the Hamming window were used to estimate the RR values. Using a window size of 32 s³⁶, our proposed method exhibits better performance than the previously used window sizes of 16 and 64 s. The signal quality index (SQI) we used is intrinsically linked to the PPG peak detector because we use SQI to extract bad signals. We used a three-point peak detector with empirically determined thresholds for PPG pulse-peak detection. For more details, see the references³⁷.

Bootstrap-based imbalanced feature generation

Generative adversarial networks (GANs) are powerful and versatile tools, especially useful for generating high-quality synthetic data and demonstrating creativity in various fields. However, training stability, expensive computational costs, and evaluation challenges can hinder our practical use. On the other hand, the main advantage of the bootstrap method is that it can provide robust, low computational cost, a simple algorithm, and flexible and reliable estimates of various statistics, especially when the underlying distribution is unknown or the sample size is small³⁸.

Handling high variance and dealing with non-linearity using bootstrap

Bootstrap is a powerful resampling technique that addresses the challenges of high variance and non-linearity in datasets in several unique ways, which generates multiple resampled datasets by randomly sampling data points from the original power spectral curves with replacement. Each resampled dataset (called a bootstrap sample) has the same size as the original power spectral curves but may contain duplicate or missing data points³¹. Generating many bootstrap samples and analyzing the variability across the results allows for estimating the sampling distribution. This process helps reduce the impact of high variance, as it accounts for how the model might behave across various subsets of the data rather than relying on a single sample³².

Bootstrap is a non-parametric method, meaning it does not make any assumptions about the functional form of the relationship between variables, which is especially useful for datasets with non-linear relationships. Instead of fitting the data to a specific linear regression, bootstrap allows for exploring the data's inherent structure by drawing resamples and evaluating performance on each. This provides a built-in method for cross-validation without requiring additional validation sets, which is particularly beneficial in high-variance scenarios where overfitting the training set might otherwise be a concern³². Therefore, the bootstrap method creates multiple independent bootstrap power spectral (PS) curves by resampling the original PS curves.

Review of power spectral (PS) features extraction

The respiratory rate (RR) estimation is calculated by multiplying the respiratory rate by 4 for a period of 15 seconds to convert it to bpm as described in¹³. Using autocorrelation functions, the respiratory frequencies can be obtained as automated feature vectors from the power spectral (PS). PS features can be automatically extracted from the resampled signals utilizing the autocorrelation function, as shown in Fig. 3a, following a preprocessing step detailed earlier. The autocorrelation function measures how similar a signal is to a delayed version of itself over successive time intervals. It is represented utilizing the variance, mean, and covariance. The average value from the input dataset is denoted by $\bar{x} = \{x_n\}_{n=1}^N$, specifying the expected value $\mathbb{E}[x]$ at each discrete time n . The mean function is represented by $\mu_x = \mathbb{E}[x]$. The autocorrelation function measures the difference between discrete times n and $n + m$. When $m = 0$ (delay 0), the autocorrelation function indicates the maximum value, representing the total energy, and is expressed as follows:

$$\nu_m(x) = \frac{\sum_{n=1}^N (x_n - \mu_x)(x_{n+m} - \mu_x)}{\sum_{n=1}^N (x_n - \mu_x)^2} \quad (1)$$

The PS represents the correlation structure of the wave signal obtained by fast Fourier transforming the correlation coefficient³⁹. The PS in the time domain exhibits autocorrelation and has the same mathematical properties as the square of the amplitude spectral. This study used a sampling frequency of 125 Hz and a record length of 400 second (s) to acquire PS for all components within the range. Conversion to a PS resulted in 50,000 signal samples for each participant. The PS curves were constructed using the autocorrelation to obtain 12×32 windows, and $12 \times 257 (=3084)$ data points were acquired from the signal (400 s) as shown in Fig. 3b. This process was repeated for the 53 subjects, resulting in $12 \times 53 (=636)$ samples and $\times 257$ feature dimensions. We included an example of the effect of PS feature curves extraction as shown in Fig. 3b.

After signal preprocessing, a clean wave signal was obtained (Fig. 1c). This clean waveform was split into segments with a 32-s window. Notably, the 32-s window segments did not overlap. The segmented waveform was converted into an autocorrelation coefficient during data analysis, and the PS was obtained from this coefficient (Fig. 3b). The PS is a mathematical tool used to analyze biological signals over time. A fast Fourier transform of the autocorrelation function should be performed to obtain the PS of all the components within 257 samples. After that, the proposed model extracts relevant features from this PS, learns them automatically, and uses them to estimate RR³⁹.

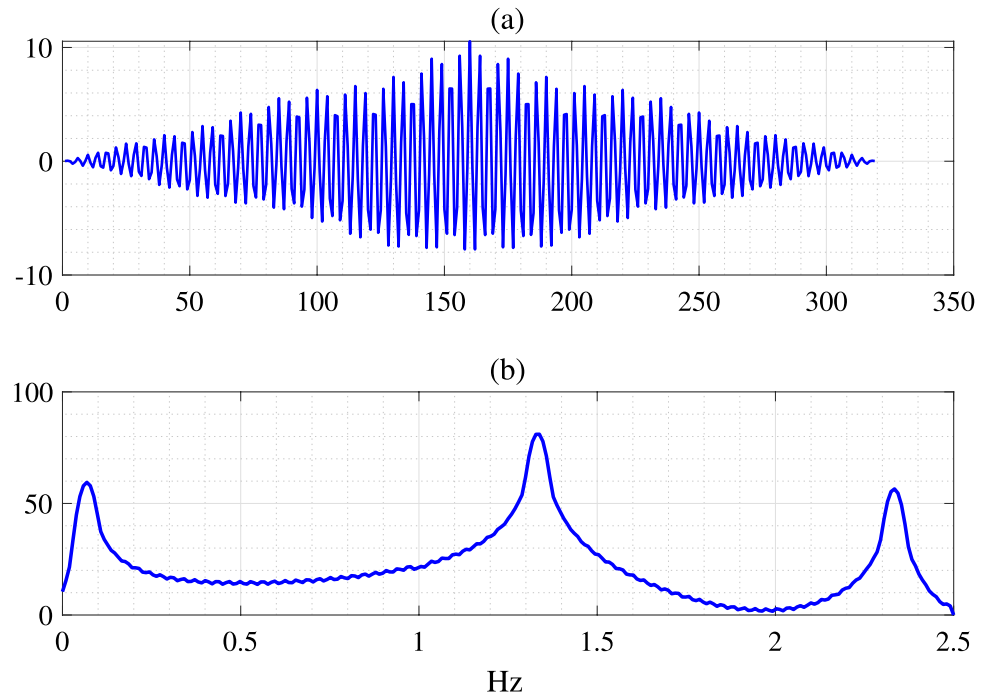


Fig. 3. (a) The upper panel is autocorrelation, and (b) the bottom panel denotes the power spectral (PS) curve from the RRSYNTH data set.

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1: procedure BIFG(P): PS curves matrix
2:    $p_\mu = \text{mean}(\mathbf{P})$ : sample estimate of mean
3:    $p_s = \text{std}(\mathbf{P})$ : sample estimate of standard deviation
4:    $p_{se} = \frac{p_s}{\sqrt{n}}$ : sample estimate of standard error
5:    $df = n - 1$ : freedom of sample
6:    $c_t = \text{tinv}(1 - \alpha/2, df)$ :  $c_t$  is estimate of the critical t-value
7:    $ci_\alpha = p_\mu - c_t \cdot p_{se}$ : estimate of lower confidence interval
8:    $ci_\beta = p_\mu + c_t \cdot p_{se}$ : estimate of upper confidence interval
9:   for  $i = 1, B$  do where B denotes the number of bootstrap sample
10:     $indx = \text{randi}(n, n, 1)$ : random resample with replacement for PS,
11:    where n denotes the number of the distribution of RR
12:    reference value (ex: < 8 bpm)
13:     $\mathbf{P}_i^* = \text{boots}(indx, :)$ :  $i$ th bootstrap sample of the PS feature
14:    matrix (curves)
15:     $p_\mu^*(i) = \text{mean}(\mathbf{P}_i^*)$ :  $i$ th bootstrap estimate of the feature's mean
16:  end for
17:   $p_s^* = \text{std}(p_\mu^*)$ : standard deviation of the bootstrap replications31
18:  for  $i = 1, B$  do
19:     $t(i) = \max(\text{abs}(p_\mu^*(i) - p_\mu) / p_s^*)$ : t-statistic
20:  end for
21:   $c^* = \text{prctile}(t, 100 \cdot (1 - \alpha/2))$ : compute test statistic at  $\alpha (= 0.05)$  level
22:   $p_\mu^* = \frac{1}{B} \sum_{i=1}^B p_\mu^*(i)$ :
23:  bias =  $p_\mu - p_\mu^*$ 
24:   $ci_\alpha^* = p_\mu^* - c^* \cdot p_s^*$ : estimate of lower confidence interval
25:   $ci_\beta^* = p_\mu^* + c^* \cdot p_s^*$ : estimate of upper confidence interval
26: end procedure

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Algorithm 1. Bootstrap-based Imbalanced Feature Generation (BIFG)

Imbalanced feature generation using bootstrap power spectral curve

Efron originally introduced the bootstrap approach in³¹. The advantage of this bootstrap method is that it does not require any modeling or assumptions regarding the data⁴⁰. In the present study, bootstrap method was used to generate artificial power curves to increase the number of imbalanced power curves.

$$P = \begin{bmatrix} p_{1,1} & \cdots & p_{1,c} \\ \vdots & \ddots & \vdots \\ p_{n,1} & \cdots & p_{n,c} \end{bmatrix} \quad (2)$$

where P denotes a PS curves measurement matrix obtained using the PS feature vector, as in (2), based on the PPG signal. n is the number of measurements, and c ($=257$) denotes the feature dimension. From the original PS curves matrix as shown Figs. 4a and 5a, B resample PS matrices were obtained $\{P_1^*, P_2^*, \dots, P_B^*\}$ using the non-parametric bootstrap approach as follows:

$$P_i^* = \begin{bmatrix} p_{1,1}^{*i} & \cdots & p_{1,c}^{*i} \\ \vdots & \ddots & \vdots \\ p_{n,1}^{*i} & \cdots & p_{n,c}^{*i} \end{bmatrix} \quad (3)$$

where P_i^* denotes in the i th artificial PS matrix acquired as shown in Figs. 4b and 5c from the original PS matrix (2) using the non-parametric bootstrap approach with random replacement of PS feature vectors.

$$p_\mu^*(i) = \frac{1}{n} \sum_{j=1}^n p_{j,\mu}^* \quad (4)$$

where $p_\mu^*(i)$ denotes i th bootstrap estimate of the mean obtained from (3). As a result, we obtain $\{p_\mu^*(1), p_\mu^*(2), \dots, p_\mu^*(B)\}$, where B ($=100$) denotes a number of bootstrap resample. Algorithm 1 provides a detailed explanation of the procedures for imbalanced feature generation and CI bands. Figure 5b denotes that the mean PS curve of the original feature, 5(c) shows the artificial PS curves using the bootstrap, and 5(d) presents the mean PS curve of the artificial feature vectors using the bootstrap approach. Figure 5e shows the difference between 5(b) and 5(d). In addition, the red line in Fig. 5f shows the standard error (SE) of the PS curves, and the blue line shows the SE of the artificial PS curves using the bootstrap. This suggests that the artificial PS curves generated using the bootstrap approach are a reliable approximation of the original power curves.

Statistical analysis of artificial feature

The two-sample Kolmogorov-Smirnov test was used to return the test results for the null hypothesis that the data of the vectors f_1 and f_2 are extracted from the same continuous distribution, where f_1 is the mean vector

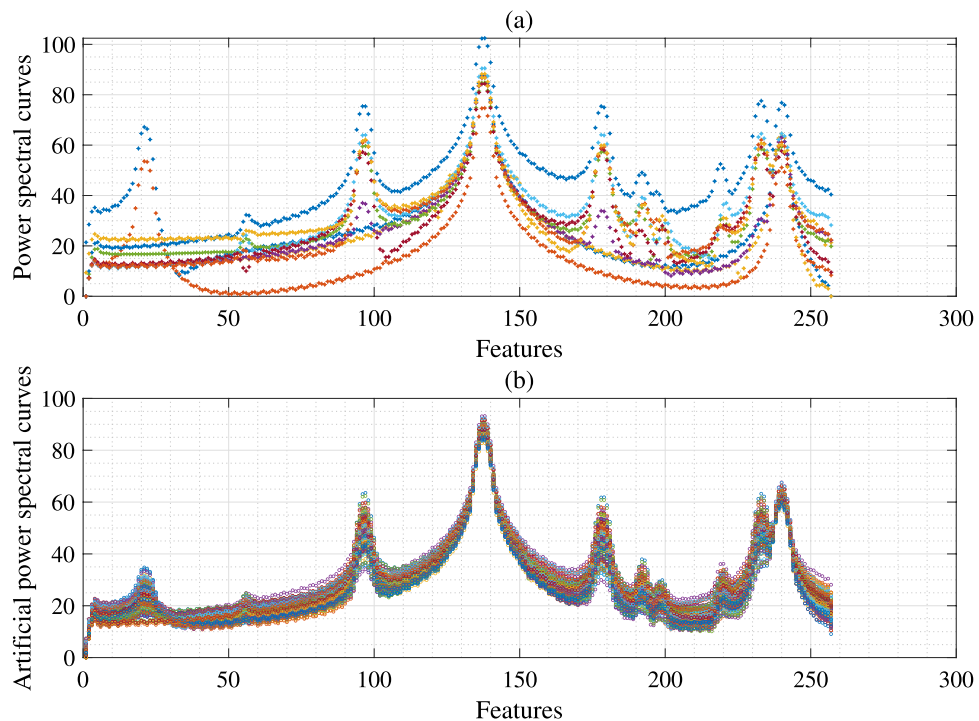


Fig. 4. (a) The upper panel is original power spectral (PS) curves from the autocorrelation (example: 24 bpm), and (b) the bottom panel denotes the artificial power spectral (PS) curves using bootstrap from the power spectral (PS) curves (a).

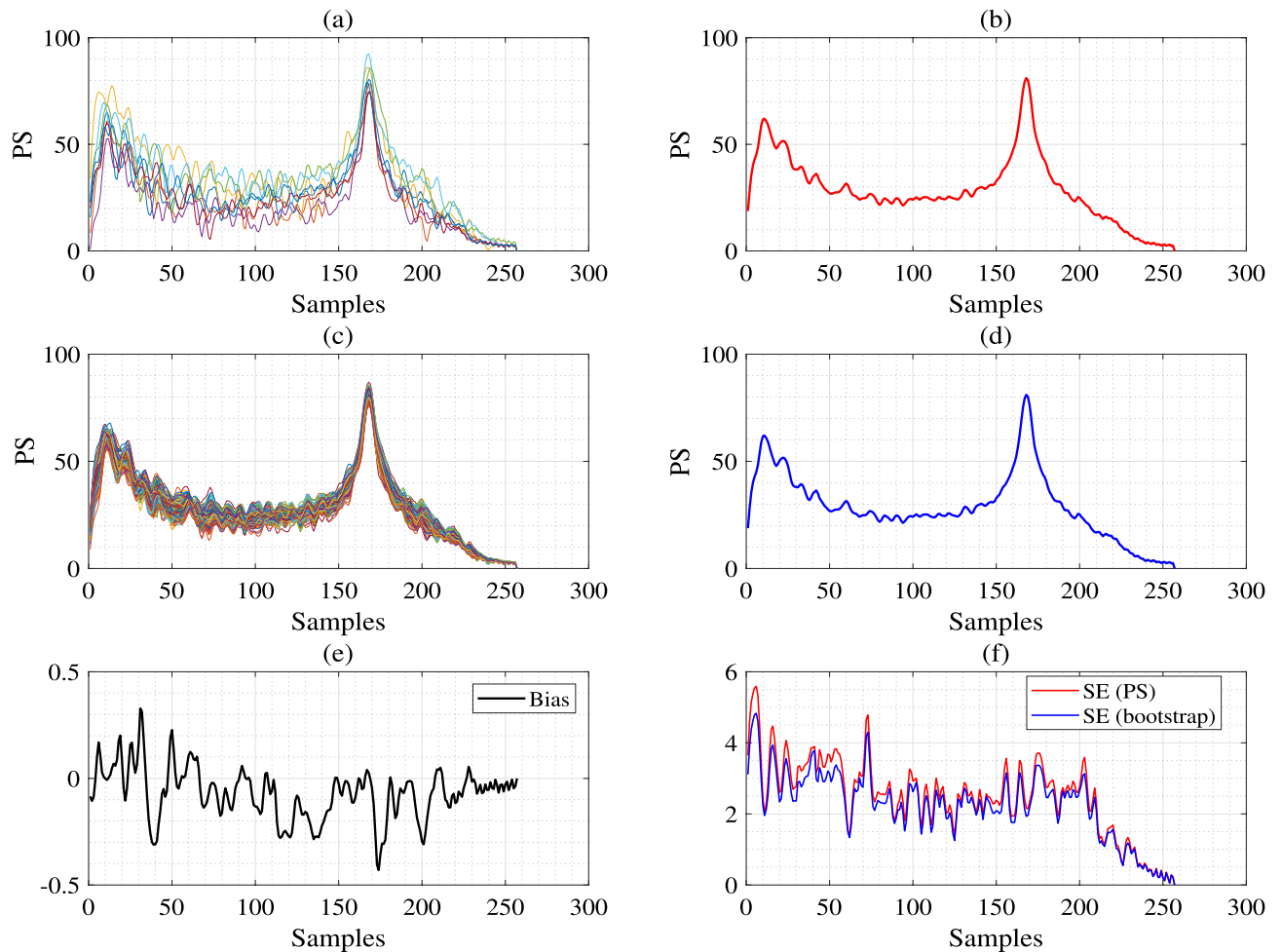


Fig. 5. Panel (a) denotes each of the PS feature curves, panel (b) is mean of the PS feature curves, panel (c) denotes each of the artificial PS feature curves using bootstrap, panel (d) is mean of the artificial PS feature curves using bootstrap, panel (e) denotes bias between mean of the PS feature curves and mean of the artificial PS feature curves using bootstrap, the red line of the panel (f) is standard error (SE) of the PS feature curves, and the blue line of the panel (f) is SE of the artificial PS feature curves using bootstrap.

for the original sample data below 8 bpm, and f_2 is the mean vector for the artificial sample data below 8 bpm. The alternative hypothesis was that f_1 and f_2 can be extracted from various continuous distributions. If the test rejected the null hypothesis at the 5% significance level, the result h was 1; otherwise, it was 0. Figure 6a and b demonstrate the cumulative distribution function (CDF) plots of the mean of the sample data (< 8) bpm and that of the sample data (> 22) bpm, respectively. The blue line denotes the mean vector of the original vector, whereas the red dashed line represents the mean vector of the artificial vector. Here, we obtained the results ($h = 0$), $p = 0.988$, and $ks = 0.016$ as shown in Fig. 6a. Therefore, we can accept the null hypothesis at the 0.05 significance level. Moreover, we confirmed that the p-values of the KS test were more than the 0.05 significance level. Furthermore, we obtained the results ($h = 0$), $p = 0.991$, and $ks = 0.023$, as shown in Fig. 6b. Therefore, we can accept the null hypothesis at the 0.05 significance level.

GPR algorithm

GPR is a powerful and flexible non-parametric method that can be utilized for supervised ML. This is beneficial when there is underlying Gaussian distribution¹⁵. First, we reviewed the Bayesian analysis of the linear model, which projects the inputs into a high-dimensional feature space and applies the linear model. The training input and output datasets are $\mathbb{D} = \{x_i, y_i\}_{i=1}^N$ and $x \in \mathcal{R}^{N \times D}$ and $y \in \mathcal{R}^{N \times 1}$, respectively. The mapping function $f = f(x)$ is used to estimate the value of y based on a given x . Depending on the criteria, GPR is employed as a non-parametric method before mapping function distribution. In addition, it is assumed that y is acquired from the corresponding $x^T w$ by adding noise as

$$y = x^T w + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I}) \quad (5)$$

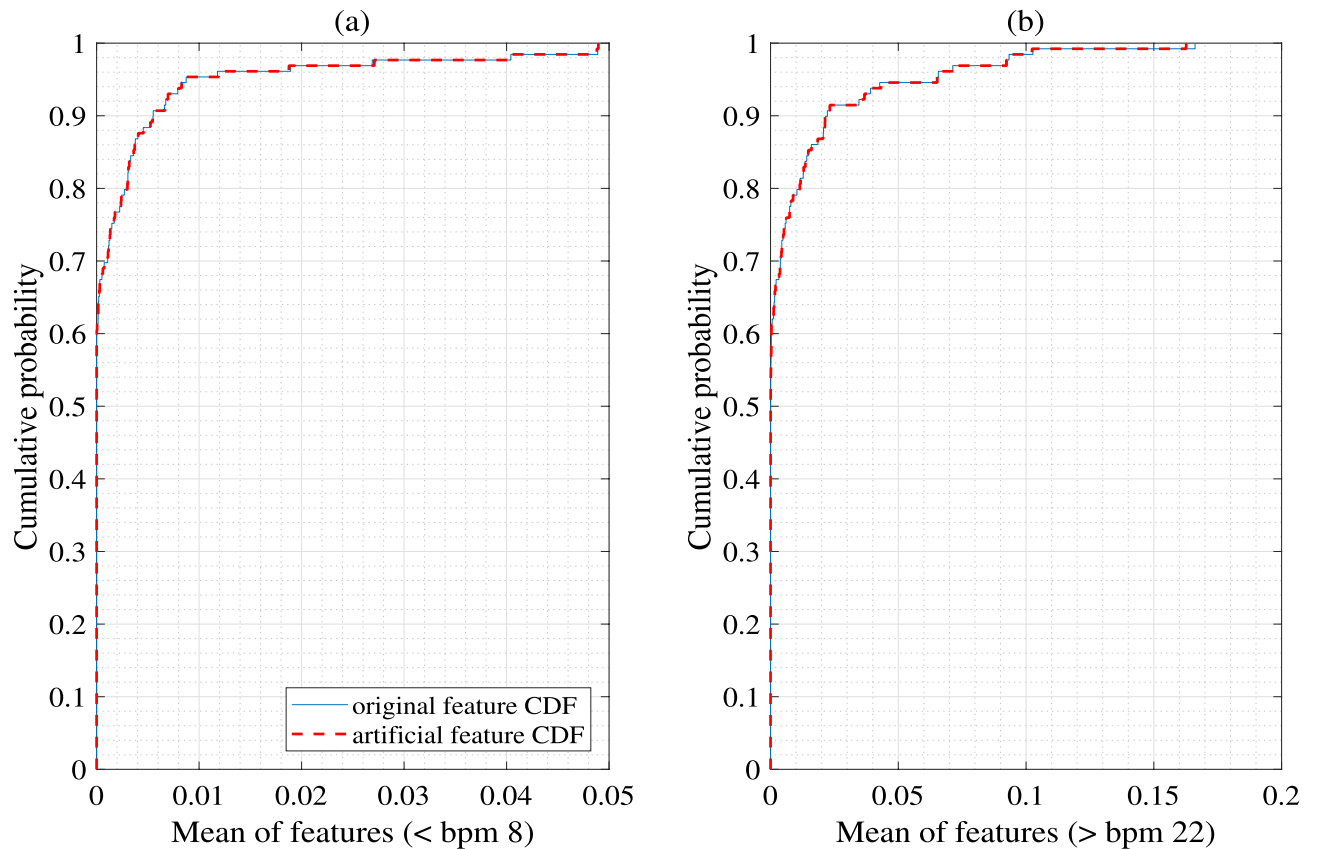


Fig. 6. Panel (a) denotes mean of feature vectors (< bpm 8), panel (b) is mean of feature vectors (> bpm 22).

Furthermore, the coefficient vector w and variance σ^2 can be obtained from the PPG signal waveform dataset. GPR predicts the target variable in the Gaussian process by including the basis η and mapping functions $f(x)$ as

$$f(x) \sim \mathbb{GP}(0, c(x, x')) \quad (6)$$

Here, $f(x)$ can be obtained from the zero-mean GPR algorithm based on the covariance $c(x, x')$ ¹⁵. In fact, the mapping can be obtained using $f(x) = \mu(x)^T w$. The mean of the data was the expected value of the mapping, denoted as $\mu(x) = E[f(x)]$. This represents the average output function for a given input value. The covariance of the latent variables determines the smoothness of the response variables. It measures the extent to which the two variables change. In addition, the basis functions were used to convert data x into a p -dimensional feature that simplifies the data by projecting it into a space where it can be analyzed and interpreted easily.

$$c(x, x') = E[(f(x) - \mu(x))(f(x') - \mu(x'))^T] \quad (7)$$

The E of (7) for the covariance is denoted as

$$c(x, x' | \theta) \approx \sigma^2 \exp\left(-\frac{\|x - x'\|^2}{2\theta^2}\right) \quad (8)$$

where c is a kernel for GPR¹⁵ and θ is a hyper-parameter. It is known that the primary hyperparameters of a typical GPR are related to the kernel function, which encodes the assumptions about the underlying function being modeled.

Currently, the most common and widely used method is maximum likelihood estimation⁴¹. Although we use cross-validation to determine hyperparameters automatically, but we briefly explain maximum likelihood estimation using the following formula. Our cross-validation for hyperparameter selection splits our fused power spectral curves into training and validation sets on the training set, and we choose the hyperparameters automatically to minimize our prediction error based on our data sets. Here, we use exponential squaring as the best-performing kernel in our study (8).

The kernel determines the properties of the mapping $f(x)$. Using the GPR, an instance of the y is expressed as

$$p(y_i|f(x_i), x_i) \sim \mathcal{N}(y_i|\eta(x_i)^T \mathbf{w} + f(x_i), \sigma^2) \quad (9)$$

where $\eta(x_i)$ represents an instance of a basis function that transforms the feature vector $\mathbf{x}$ into a new feature vector $\eta(\mathbf{x})$. Thus, we can define $\boldsymbol{\Omega} = \{\mathbf{w}, \theta, \sigma^2\}$ from the data set $\mathbb{D}$ and the marginal likelihood as

$$p(y|\mathbf{x}) = p(y|\mathbf{x}, \boldsymbol{\Omega}) \approx \mathcal{N}(y|\boldsymbol{\Theta}\mathbf{w}, c(\mathbf{x}, \mathbf{x}'|\theta) + \sigma^2\mathbf{I}), \quad (10)$$

GPR training and suitable kernel selection were based on specific hypotheses, including the expected patterns and smoothness of the data. The logarithmic marginal likelihood was maximized to estimate the hyperparameter $\boldsymbol{\Omega}$ as

$$\log p(y|\mathbf{x}, \boldsymbol{\Omega}) = -\frac{1}{2} \log |c(\mathbf{x}, \mathbf{x}'|\theta) + \sigma^2\mathbf{I}| - \frac{1}{2} n \log 2\pi - \frac{1}{2} (\mathbf{y} - \boldsymbol{\Theta}\mathbf{w})^T [c(\mathbf{x}, \mathbf{x}'|\theta) + \sigma^2\mathbf{I}]^{-1} (\mathbf{y} - \boldsymbol{\Theta}\mathbf{w}) \quad (11)$$

where $c(\mathbf{x}, \mathbf{x}'|\eta)$ is the covariance matrix and $\boldsymbol{\Theta}$ denotes the basis matrix. The log-marginal likelihood is a penalty fit scale used to maximize the probability of a particular model using a gradient ascent optimization technique. This technique involves iteratively adjusting model parameters to maximize the log-marginal likelihood until convergence is achieved. Therefore, we can obtain the best possible model fit that maximizes the probability of the observed data.

The hyper-parameters $\boldsymbol{\Omega} = \{\mathbf{w}, \theta, \sigma^2\}$ using the GPR is by maximizing the likelihood $p(y|\mathbf{x})$ as a function for $\boldsymbol{\Omega}$.

$$\mathbb{L}(\hat{\boldsymbol{\Omega}}) = \arg \max_{\boldsymbol{\Omega}} \log(y|\mathbf{x}, \boldsymbol{\Omega}) \quad (12)$$

Here, we were concerned about some statistical calculations associated with hyperparameter estimation. In particular, the values of $\hat{\mathbf{w}}(\theta, \sigma^2)$ were calculated to maximize the log-likelihood function for a given hyperparameter set (θ, σ^2) . This is a general approach for statistical inference to estimate the value of unknown parameters based on the observed data.

$$\hat{\mathbf{w}}(\theta, \sigma^2) = \left\{ \boldsymbol{\Theta}^T [c(\mathbf{x}, \mathbf{x}'|\theta) + \sigma^2\mathbf{I}]^{-1} \boldsymbol{\Theta} \right\}^{-1} \boldsymbol{\Theta}^T [c(\mathbf{x}, \mathbf{x}'|\theta) + \sigma^2\mathbf{I}]^{-1} \mathbf{y} \quad (13)$$

To make probabilistic estimations for the Bayesian GPR model using known hyperparameters, we may need to use $p(y^*|y, \mathbf{x}, \mathbf{x}^*)$. However, if we have a finite input of new data $\mathbf{x}^*$, we can still estimate the target y using multivariate normal distributions. For these distributions, the covariance matrices will be generated by the kernel and used to estimate the output of the new data. Therefore, the conditional distribution was determined as follows:

$$p(y^*|y, \mathbf{x}, \mathbf{x}^*) = \frac{p(y^*, y|\mathbf{x}, \mathbf{x}^*)}{p(y|\mathbf{x}, \mathbf{x}^*)} \quad (14)$$

The mapping functions f^* and f is used to acquire the joint probability in the numerator, such as (14)

$$\begin{aligned} p(y^*, y|\mathbf{x}, \mathbf{x}^*) &= \int \int p(y^*, y, f^*, f|\mathbf{x}, \mathbf{x}^*) df df^* \\ &= \int \int p(y^*, y|f^*, f, \mathbf{x}, \mathbf{x}^*) p(f^*, f|\mathbf{x}, \mathbf{x}^*) df df^* \end{aligned} \quad (15)$$

The GPR assumes that each target value (represented as y_i) is dependent on both the corresponding latent variable ($f(x_i)$) and the feature (x_i). Based on the given $y, \mathbf{x}$ and hyperparameters $\boldsymbol{\Omega}$, the model estimates the expected value of the estimation.

$$\begin{aligned} E(y^*|y, \mathbf{x}, \mathbf{x}^*, \boldsymbol{\Omega}) &= \eta(\mathbf{x}^*)^T \mathbf{w} + c(\mathbf{x}, \mathbf{x}'|\theta) \varphi \\ &= \eta(\mathbf{x}^*)^T \mathbf{w} + \sum_{i=1}^N \varphi_i c(\mathbf{x}^*, x_i|\theta) \end{aligned} \quad (16)$$

Here, $\varphi = [c(\mathbf{x}, \mathbf{x}) + \sigma^2\mathbf{I}]^{-1} (\mathbf{y} - \boldsymbol{\Theta}\mathbf{w})$. In fact, we can define an optimal point-estimation $\hat{\mathbf{y}}^*$ based on a cost as

$$E_{\mathbb{C}}(\hat{\mathbf{y}}^*|\mathbf{x}^*) = \int \mathbb{C}(y^*, \hat{\mathbf{y}}^*) p(y^*|\mathbf{x}^*, \mathbb{D}) dy^* \quad (17)$$

The target is to acquire an estimate $y^* \approx \hat{\mathbf{y}}^*$ and minimize the expected value of the cost that $\mathbb{C}(y^*, \hat{\mathbf{y}}^*)$ is calculated by minimizing between y^* and $\hat{\mathbf{y}}^*$ as

$$\hat{\mathbf{y}}_{\text{optimal}}|\mathbf{x}^* = \arg \min_{\hat{\mathbf{y}}^*} E_{\mathbb{C}}(\hat{\mathbf{y}}^*|\mathbf{x}^*) \quad (18)$$

We are familiar with the mean absolute error (MAE), a popular metric used to evaluate the accuracy of estimations. The MAE measures the average amount by which the estimated values differ from the actual values in absolute terms.

Experimental results

The parameters used in the proposed and conventional methods were tuned using the training stage. As mentioned in the introduction, machine learning models such as SVM¹⁴, NN¹⁶, gradient boosting algorithm (GBA)¹¹, LSTM¹³, and GPR¹⁵ were used to compare the performance with the proposed BIFG with GPR, which have been popular in continuous RR estimation until recently. The SVM model¹⁴ was suitable for handling nonlinear relationships between features extracted from PPG signals and reference RR, as it can handle nonlinear relationships in input and output data well. Many researchers have used the NN models for medical application^{16,22}. Different deep learning models have recently been used for RR estimation based on PPG signals^{12,13}.

The tuning stage of parameters was crucial because it can considerably improve the performance of machine learning models. This study's five-fold cross-validation fine-tuned the parameters for the conventional methods and GPR¹⁵. The main parameters for each model were defined, and the possible range of values was determined for each parameter, highlighting our research's potential benefits and advancements. The process began with conventional methods and GPR models, which conducted a grid search for all possible parameter combinations to identify the parameters for achieving the best performance results. Subsequently, this five-fold cross-validation was performed to enhance the robustness of the ML models using the optimal parameters identified. The training data were then divided into five non-overlapping subsets of equal sizes for iterative learning. Each iteration involved four-fold training and assessment of the model using the remaining folds.

After preprocessing, the resampled wave signals were generated to remove noise and outliers from the PPG signals. The proposed method was generated using the 53 and 192-long resampled wave signals, respectively. In the first experimental scenario, the PPG signals were sequentially split into 80% and 20% for the training dataset to avoid mixing training and test data. Subsequently, the reference RR values were calculated from the oral and nasal pressure signals using a custom respiration detection algorithm³⁴. In the second experimental scenario, the PPG signals were sequentially divided into a training set (80%), a validation set (10%), and a test set (10%).

Table 1 shows the parameter adjustments for the conventional and proposed BIFG models using PPG signals based on the RRSYNTH data set³⁴. Parameter tuning was omitted when using the BIDMC dataset to save paper space since the RRSYNTH dataset is more extensive. MATLAB 2023⁴² was used to measure the execution (training and testing) time based on the RRSYNTH dataset listed in Table 2. The test time measurement results showed that the computational complexity of each benchmark model and BIFG combination was almost the same, so it can be expected that there will be no problem in applying it to a real home-based monitoring system, as shown in Table 2.

From the data set, we acquired $12 \times 53 (= 636)$ samples with 257 feature dimensions, where 53 represents the number of patients. Furthermore, we obtained 7 $\leq$, 8 $\geq$ and 9 $\leq$, 10 $\geq$ and 11 $\leq$, 22 $\geq$ and 23 $\leq$, 24 $\geq$ and 25 $\leq$, and 26 $\geq$ bpm samples from 636 original samples, used them as input data for the proposed BIFG algorithm, and obtained 600 artificial samples using the proposed method as shown in Fig. 4b. Therefore, we finally obtained 1236 samples with 257 feature dimensions based on the BIDMC data set. We also obtained $6 \times 192 (= 1152)$ samples with 257 feature dimensions, where 192 denotes the number of records, and combined 1152 samples with 600 artificial samples using the proposed BIFG method. Hence, we acquired 1752 samples using the RRSYNTH data set. Because of space limitations in the paper, the results of the second experimental scenario are presented in Tables 3, 4, 5, and 6. We will omit the detailed analysis and discussion. Overall, the evaluation results are similar to those of the first scenario experiment regarding mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R^2). The revised results confirm that our model performs consistently well with the new split and does not overfit based on the second experimental scenario.

Table 3 shows the performance evaluation results of GPR by applying the BIFG algorithm to four benchmark algorithms: SVM¹⁴, NN¹⁶, GBA¹¹, and LSTM¹³. For the objective performance evaluation of the BIFG algorithm proposed in this table, the results obtained by combining BIFG with all learning algorithms and without combining the BIFG algorithm were evaluated using the public BIDMC and RRSYNTH datasets. The evaluation used MAE and standard deviation (SD) using PPG-based PS curve features. Each feature set was inputted into each ML model, and the resulting MAE and SD values were recorded.

The average of 30 iterations showed that the SVM-BIFG algorithm (MAE 1.82 bpm) outperforms the SVM algorithm (MAE 1.99 bpm). Here, the difference in MAE between the SVM model and the combined SVM-BIFG model shows that the MAE is reduced by 0.17 bpm. Comparing the performance difference using the R-squared in Table 5, the SVM acquired 0.63, and the combined SVM-BIFG model obtained 0.94. This indicates that a 0.17 bpm reduction in MAE leads to a 0.31 (31%) performance improvement in R-squared. The 0.17 bpm reduction clearly shows that the performance improvement in RR is simply $9.3\% = (1.99 - 1.82) / 1.82 \times 100$ through performance improvement calculation. The NN-BIFG and the conventional NN algorithms showed MAE values of (1.42 bpm and 2.15 bpm), respectively, indicating that the proposed BIFG-combined NN algorithm also showed superior performance. This represents that an MAE reduction of 0.73 bpm is accompanied by an R-squared increase of 0.37 (37%) in Table 5. The 0.73 bpm reduction shows that the performance improvement in RR is simply $51.4\% = (2.15 - 1.42) / 1.42 \times 100$. The GBA-BIFG algorithm performed better (MAE 1.20 bpm) than the GBA algorithm (MAE 1.80 bpm). This represents that an MAE reduction of 0.60 bpm is accompanied by an R-squared increase of 0.36 (36%) in Table 5. The 0.60 bpm reduction denotes that the performance improvement in RR is simply $50\% = (1.80 - 1.20) / 1.20 \times 100$.

Methods	SVM ¹⁴	NN ¹⁶	GBA ¹¹	LSTM ¹³	GPR ¹⁵
Combined	BIFG	BIFG	BIFG	BIFG	BIFG
Data set : RRSYNTH					
Number of Training Sample	1402	1402	1402	1402	1402
Number of Testing Sample	350	350	350	350	350
Number of Feature	257	257	257	257	257
Output Dimension	1	1	1	1	1
Learner			tree		
Shrinkage Factor			0.1		
Max Tree Depth			uint32(6)		
Rand Seed			uint32(rand()*1000)		
Number of Iteration			1000		
KFold	5			5	5
Layer Size		10			
Activations		Relu		Relu	
Lambda		0			
Standardize	False	False		False	
Optimizer					Quasi newton
Box Constraint					
Epsilon	3.0				
Kernel Scale	auto				
Solver	ISDA			adam	
Kernel Function	gaussian				exponential
Basic Function					none
Fit Method					exact
Predict Method					exact
Active Set Method					random
Number of HiddenUnits				500	
Layer				Sequence Input Layer	
				Lstm Layer	
				Fully Connected Layer (50)	
				Dropout Layer (0.5)	
				Fully Connected Layer (1)	
				Regression Layer	
Max Epochs				200	
Initial Learn Rate				0.01	
Mini Batch Size				1	
Gradient Threshold				1	
Drop Factor				0.5	
Shuffle				Never	
Loss			Squaredloss	MSE	
Sequence Length				Longest	
Metrics				RMSE	
Verbose	0	0		0	0

Table 1. Summary parameters for conventional and the proposed GPR with BIFG methods using the public RRSYNTH data set³⁴. These results were acquired using the first experimental scenario.

In addition, the LSTM-BIFG and LSTM models recorded MAE values of (1.70 bpm and 2.49 bpm), respectively, showing that the LSTM-BIFG combined method showed superior performance. This shows that an MAE reduction of 0.79 bpm is accompanied by an R-squared increase of 0.40 (40%) in Table 5. The 0.79 bpm reduction indicates that the performance improvement in RR is simply $46.5\% = (2.49 - 1.70) / 1.70 \times 100$. The proposed GPR-BIFG algorithm achieved better (MAE 0.89 bpm) result than the GPR (MAE 1.45 bpm), this represents that an MAE reduction of 0.56 bpm is accompanied by an R-squared increase of 0.20 (20%) in Table 5. Here, the 0.20 bpm reduction denotes that the performance improvement in RR is simply $62.9\% = (1.45 - 0.89) / 0.89 \times 100$ based on the BIDMC data set. The SD values of the conventional and proposed models showed stable results.

Using the RRSYNTH data set, the SVM-BIFG algorithm performed better (MAE 3.51 bpm) than the SVM algorithm (MAE 4.92 bpm). This shows that an MAE reduction of 1.41 bpm is accompanied by an R-squared

Methods	SVM	SVM	NN	NN	GBA	GBA	LSTM	LSTM	GPR	GPR
Combined		BIFG		BIFG		BIFG		BIFG		BIFG
Training time (s)	0.17	0.09	2.19	2.11	2.24	3.28	23.12	29.40	1.43	2.35
Testing time (s)	0.02	0.01	0.01	0.01	0.03	0.03	0.28	0.11	0.03	0.02

Table 2. Comparing training and testing times between the conventional and proposed methodologies using Intel Core i5-9400 CPU 4.1 GHz, OS 64 bit, RAM 16.0 GB, and Matlab 2023 (The MathWorks Inc., Natick, Ma, USA) system specifications based on the RRSYNTH data set, where BIFG denotes the proposed bootstrap-based imbalanced feature generation (BIFG) model. These results were acquired using the first experimental scenario.

Data set	Error (bpm)	SVM	SVM	NN	NN	GBA	GBA	LSTM	LSTM	GPR	GPR
	Combined		BIFG		BIFG		BIFG		BIFG		BIFG
BIDMC	MAE (1st)	1.99	1.82	2.15	1.42	1.80	1.20	2.14	1.70	1.45	0.89
	SD (1st)	0.12	0.15	0.16	0.23	0.11	0.07	0.37	0.28	0.10	0.06
	MAE (2nd)	1.58	1.01	1.71	0.86	1.81	1.20	2.26	2.13	1.43	0.82
	SD (2nd)	0.14	0.07	0.13	0.06	0.15	0.09	0.40	0.36	0.11	0.07
RRSYNTH	MAE (1st)	4.92	3.51	3.90	3.22	5.05	3.84	5.00	4.32	2.19	1.44
	SD (1st)	0.39	0.27	0.88	1.06	0.35	0.22	0.45	0.46	0.21	0.14
	MAE (2nd)	4.82	3.37	4.02	3.53	4.99	4.01	5.12	4.41	2.30	1.52
	SD (2nd)	0.36	0.45	0.41	0.32	0.36	0.27	0.85	0.66	0.32	0.25

Table 3. Using the PPG-based multiple algorithms, are computed as the difference from the reference RR values to express it as the MAE and SD results. Where 1st is the first experimental scenario, 2nd denotes the second experimental scenario.

Data set	Error (bpm)	SVM	SVM	NN	NN	GBA	GBA	LSTM	LSTM	GPR	GPR
	Combined		BIFG		BIFG		BIFG		BIFG		BIFG
BIDMC	RMSE (1st)	2.31	1.97	2.48	2.11	2.41	1.84	2.52	1.98	1.93	1.52
	SD (1st)	0.14	0.16	0.19	0.27	0.21	0.14	0.23	0.15	0.15	0.11
	RMSE (2nd)	2.00	1.48	2.24	1.43	2.41	1.77	2.59	2.43	1.93	1.43
	SD (2nd)	0.19	0.10	0.19	0.11	0.24	0.15	0.25	0.20	0.16	0.11
RRSYNTH	RMSE (1st)	8.40	5.46	6.04	5.26	8.12	6.55	7.70	6.22	3.94	3.30
	SD (1st)	0.75	0.79	1.11	1.44	0.65	0.57	0.63	0.55	0.46	0.50
	RMSE (2nd)	8.33	5.24	6.12	5.48	8.05	6.69	7.84	6.30	4.02	3.34
	SD (2nd)	0.57	0.36	0.54	0.42	0.70	0.62	0.72	0.48	0.41	0.48

Table 4. Using the PPG-based multiple algorithms are computed as the difference from the reference RR values to express it as the RMSE and SD results.

Data set	Error (bpm)	SVM	SVM	NN	NN	GBA	GBA	LSTM	LSTM	GPR	GPR
	Combined		BIFG		BIFG		BIFG		BIFG		BIFG
BIDMC	R ² (1st)	0.63	0.94	0.56	0.93	0.59	0.95	0.54	0.94	0.76	0.96
	SD (1st)	0.08	0.01	0.05	0.03	0.05	0.01	0.05	0.01	0.06	0.01
	R ² (2nd)	0.76	0.97	0.68	0.96	0.62	0.96	0.54	0.92	0.78	0.97
	SD (2nd)	0.04	0.02	0.04	0.01	0.06	0.01	0.06	0.01	0.04	0.01
RRSYNTH	R ² (1st)	0.38	0.79	0.71	0.76	0.44	0.69	0.52	0.72	0.90	0.93
	SD (1st)	0.07	0.05	0.11	0.26	0.09	0.04	0.09	0.04	0.02	0.02
	R ² (2nd)	0.39	0.79	0.72	0.75	0.45	0.67	0.40	0.67	0.89	0.93
	SD (2nd)	0.09	0.06	0.15	0.24	0.09	0.05	0.08	0.05	0.03	0.03

Table 5. Using the PPG-based multiple algorithms are computed as the difference from the reference RR values to express it as the R² and SD results.

Methods & Combined	GPR	GPR BIFG	GPR	GPR BIFG
Data set	BIDMC	BIDMC	RRSYNTH	RRSYNTH
RR (SD) (1st)	17.45 (0.22)	16.24 (0.38)	20.88 (0.38)	19.11 (0.37)
RR (SD) CI L (1st)	13.47 (0.25)	13.73 (0.33)	10.70 (0.50)	11.89 (0.48)
RR (SD) CI U (1st)	21.43 (0.25)	18.74 (0.48)	31.05 (0.50)	26.32 (0.53)
RR (SD) 95% CI (1st)	7.96 (0.27)	5.01 (0.24)	20.35 (0.65)	14.43 (0.69)
RR (SD) (2nd)	17.40 (0.21)	16.65 (0.27)	21.12 (0.38)	19.10 (0.30)
RR (SD) CI L (2nd)	13.42 (0.23)	14.50 (0.31)	10.63 (0.47)	11.59 (0.52)
RR (SD) CI U (2nd)	21.39 (0.25)	18.79 (0.27)	31.61 (0.49)	26.62 (0.49)
RR (SD) 95% CI (2nd)	7.97 (0.22)	4.29 (0.21)	20.98 (0.86)	15.03 (0.81)

Table 6. We compare the CIs obtained using the GPR and GPR-BIFG, where U denotes upper limit and L denotes lower limit.

increase of 0.41 (41 %) in Table 5. Thus, the 1.41 bpm reduction is that the performance improvement in RR is simply $40.2\% = (4.92 - 3.51) / 3.51 \times 100$. The NN-BIFG and the existing NN algorithm showed MAE values (3.22bpm and 3.90bpm), respectively, indicating that the proposed BIFG combined NN algorithm also showed slightly better performance. The difference between the two methods was 0.68 bpm, and this 0.68 bpm performance improvement is accompanied by an R-squared increase of 0.02 (2.0 %) in Table 5. Here, the 0.68 bpm reduction is that the performance improvement in RR is simply $21.1\% = (3.90 - 3.22) / 3.22 \times 100$. The GBA-BIFG algorithm performed better (MAE 3.84 bpm) than the GBA algorithm (MAE 5.05 bpm). This represents that an MAE reduction of 1.21 bpm is attended by an R-squared increase of 0.41 (41 %) in Table 5. Hence, the 1.21 bpm reduction denotes that the performance improvement in RR is simply $31.5\% = (5.05 - 3.84) / 3.84 \times 100$.

The LSTM-BIFG and LSTM algorithms also obtained MAE values of (4.32 bpm and 5.00 bpm), respectively, showing that the LSTM-BIFG combined method represented slightly better performance. This shows that an MAE reduction of 0.68 bpm is acquired by an R-squared increase of 0.20 (20 %) in Table 5. Hence, the 0.68 bpm reduction indicates that the performance improvement in RR is simply $15.7\% = (5.00 - 4.32) / 4.32 \times 100$.

The proposed GPR-BIFG algorithm (MAE 1.44 bpm) achieved better results than the GPR algorithm (MAE 2.19 bpm), as shown in Table 3. This indicates that reducing the MAE by 1.21 bpm in Table 5 increases the R-squared by 0.03 (3.0 %). Therefore, a 1.21 bpm reduction implies a performance improvement of RR of $52.1\% = (2.19 - 1.44) / 1.44 \times 100$. Here, the first thing to note is that a 1.21 bpm reduction due to the BIFG algorithm combined with the performance improvement implies a performance improvement of 52.1 % in healthcare systems. The second thing to note is that the MAE difference between NN (MAE 3.90 bpm) and NN-BIFG (MAE 3.22 bpm) is 0.68 bpm, while the MAE difference between LSTM (5.0 bpm) and LSTM-BIFG (4.32 bpm) is 0.68 bpm. Although the MAE decreases by the same 0.68 bpm, when considering the aspect of system performance improvement, combining NN and BIFG shows a performance improvement of $21.1\% = (3.90 - 3.22) / 3.22 \times 100$, while combining LSTM and the proposed BIFG shows a performance improvement of $15.7\% = (5.00 - 4.32) / 4.32 \times 100$.

Based on the results presented in Table 4, both the existing and proposed algorithms were evaluated based on the root mean square error (RMSE) and SD values. The SVM-BIFG model had a lower RMSE value (1.97 bpm) than the SVM model (2.31 bpm), which indicates that the RMSE was reduced by 0.34 bpm. Therefore, a reduction of 0.34 bpm means that the performance improvement of RR is simply $17.3\% = (2.31 - 1.97) / 1.97 \times 100$ with respect to the performance of the home-based healthcare system. Similarly, the NN-BIFG model had a lower RMSE value (2.11 bpm) than the NN model (2.48 bpm), which shows that the RMSE was reduced by 0.37 bpm. Thus, a reduction of 0.37 bpm means that the performance improvement of RR is simply $17.5\% = (2.48 - 2.11) / 2.11 \times 100$ with respect to the performance of the home-based healthcare system.

In addition, the GBA-BIFG model outperformed the conventional GBA model, with RMSE values of 1.84 bpm and 2.41 bpm, respectively. This difference indicates that the RMSE was reduced by 0.57 bpm. Therefore, a reduction of 0.57 bpm means that the performance improvement of RR is simply $31\% = (2.41 - 1.84) / 1.84 \times 100$ with respect to the performance of the healthcare monitoring system. The RMSE value of the LSTM-BIFG model (1.98 bpm) was lower than that of the LSTM model (2.52 bpm), which shows that the RMSE was reduced by 0.54 bpm. Therefore, a reduction of 0.54 bpm means that the performance improvement of RR is simply $27.3\% = (2.52 - 1.98) / 1.98 \times 100$ with respect to the performance of the healthcare monitoring system. Notably, among all the evaluated models, including the GPR model (1.93 bpm), the proposed GPR-BIFG model had the lowest RMSE value (1.52 bpm). This difference shows that the RMSE was reduced by 0.41 bpm. Hence, a reduction of 0.41 bpm means that the performance improvement of RR is simply $27\% = (1.93 - 1.52) / 1.52 \times 100$ based on the BIDMC data set with respect to the performance of the healthcare monitoring system.

For RMSE based on the RRSYNTH data set, the SVM-BIFG model obtained a lower RMSE value (5.46 bpm) than the SVM model (8.40 bpm). The difference between these two models is 2.94 bpm, which indicates a $53.8\% = (8.40 - 5.46) / 5.46 \times 100$ performance improvement in terms of the healthcare monitoring system. The NN-BIFG model obtained a lower RMSE value (5.26 bpm) than the NN model (6.04 bpm). The difference between these two models shows 0.78 bpm, which denotes a $14.8\% = (6.04 - 5.26) / 5.26 \times 100$ performance improvement in terms of the healthcare monitoring system. The GBA-BIFG model outperformed the conventional GBA model, with RMSE values of 6.55 bpm and 8.12 bpm, respectively. The difference between these two models is 1.57 bpm, which indicates a $24.0\% = (8.12 - 6.55) / 6.55 \times 100$ performance improvement in terms of the

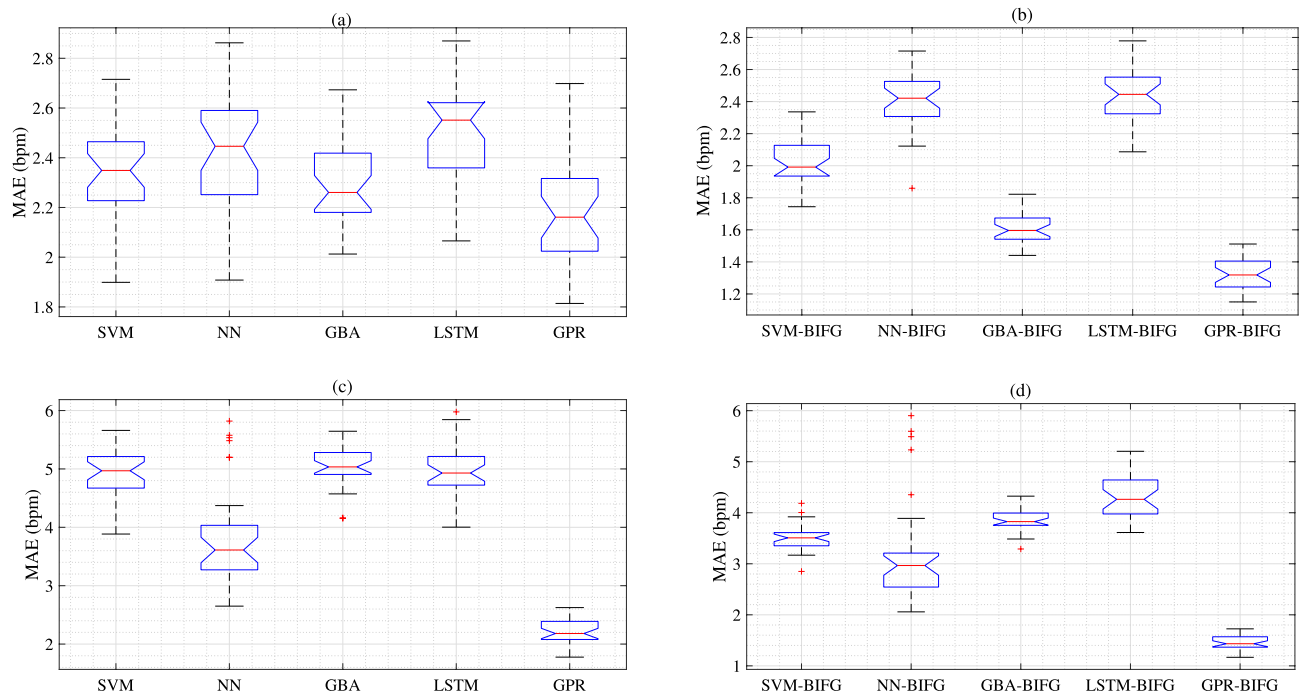


Fig. 7. Panel (a) represents the MAE and SD compared to the reference RR method acquired from the SVM, NN, GBA, LSTM, and GPR algorithms based on the BIDMC data set. Panel (b) shows the MAE and SD compared to the reference RR method obtained from the SVM-BIFG, NN-BIFG, GBA-BIFG, LSTM-BIFG, and GPR-BIFG algorithms based on the BIDMC data set. Panel (c) represents the MAE and SD compared to the reference RR method acquired from the SVM, NN, GBA, LSTM, and GPR algorithms based on the RRSYNTH data set. Panel (d) shows the MAE and SD compared to the reference RR method obtained from the SVM-BIFG, NN-BIFG, GBA-BIFG, LSTM-BIFG, and GPR-BIFG algorithms based on the RRSYNTH data set. These results were acquired using the first experimental scenario.

healthcare monitoring system. The RMSE value of the LSTM-BIFG model (6.22 bpm) was lower than that of the LSTM model (7.70 bpm). The difference between these two models is 1.48 bpm, which denotes a 23.8% $= (7.70 - 6.22) / 6.22 \times 100$ performance improvement in terms of the healthcare monitoring system. The proposed GPR-BIFG model obtained the lowest RMSE value (3.30 bpm), including the GPR model (3.94 bpm). The difference between these two models denotes 1.57 bpm, which indicates a 19.4% $= (3.94 - 3.30) / 3.30 \times 100$ performance improvement in terms of the healthcare monitoring system.

Figure 7a presented the MAE values of the SVM, NN, GBA, LSTM, and GPR algorithms compared with the reference RR values listed in Table 3. The GPR model had a lower MAE value than the SVM, NN, GBA, and LSTM models. Furthermore, statistical analysis revealed that the GPR algorithm significantly differed from the other models, with a multiple comparison p -value of $(1.47e - 7)$. Figure 7b illustrates the MAE values of the SVM-BIFG, NN-BIFG, GBA-BIFG, LSTM-BIFG, and GPR-BIFG models concerning the reference RR values. The results indicate a significant difference between the GPR-BIFG and other models, with a multiple comparison p -value of $(8.07e - 75)$ based on the BIDMC data set.

Using the the RRSYNTH data set, the MAE values of the SVM, NN, GBA, LSTM, and GPR algorithms were compared with the reference RR values listed in Table 3 as shown in Fig. 7c. The GPR model had a lower MAE value than the SVM, NN, GBA, and LSTM models. Moreover, statistical analysis denoted that the GPR algorithm significantly differed from the other models, with a multiple comparison p -value of $(2.12e - 54)$. Figure 7d presented the MAE values of the SVM-BIFG, NN-BIFG, GBA-BIFG, LSTM-BIFG, and GPR-BIFG models concerning the reference RR values. The results indicate a significant difference between the GPR-BIFG and other models, with a multiple comparison p -value of $(1.05e - 45)$ based on the SYNTH data set.

The R^2 is a statistical measure that ranges from 0 to 1. Values closer to 1 indicate a stronger relationship between predictor and response variables. Table 5 showed that the (SVM-BIFG:0.94) model had a stronger correlation with the response compared to the (SVM:0.63) model. On the other hand, the R^2 of the (NN:0.56) model indicated a not stronger correlation with the response variable when compared to the (NN-BIFG:0.93) model. Similarly, the R^2 of the (GBA-BIFG:0.95) model indicated a higher correlation with the response variable than that of the (GBA:0.59) model. However, the R^2 of the (LSTM:0.54) model indicated low correlations with the response variables compared to the (LSTM-BIFG:0.94) model as presented in Fig. 8c and d. In fact, the R^2 of the (GPR-BIFG: 0.96) model was the highest of all the models, indicating the strongest correlation with the response variable than that of the (GPR:0.76) model based on the BIDMC data set, as shown in Fig. 8a and b.

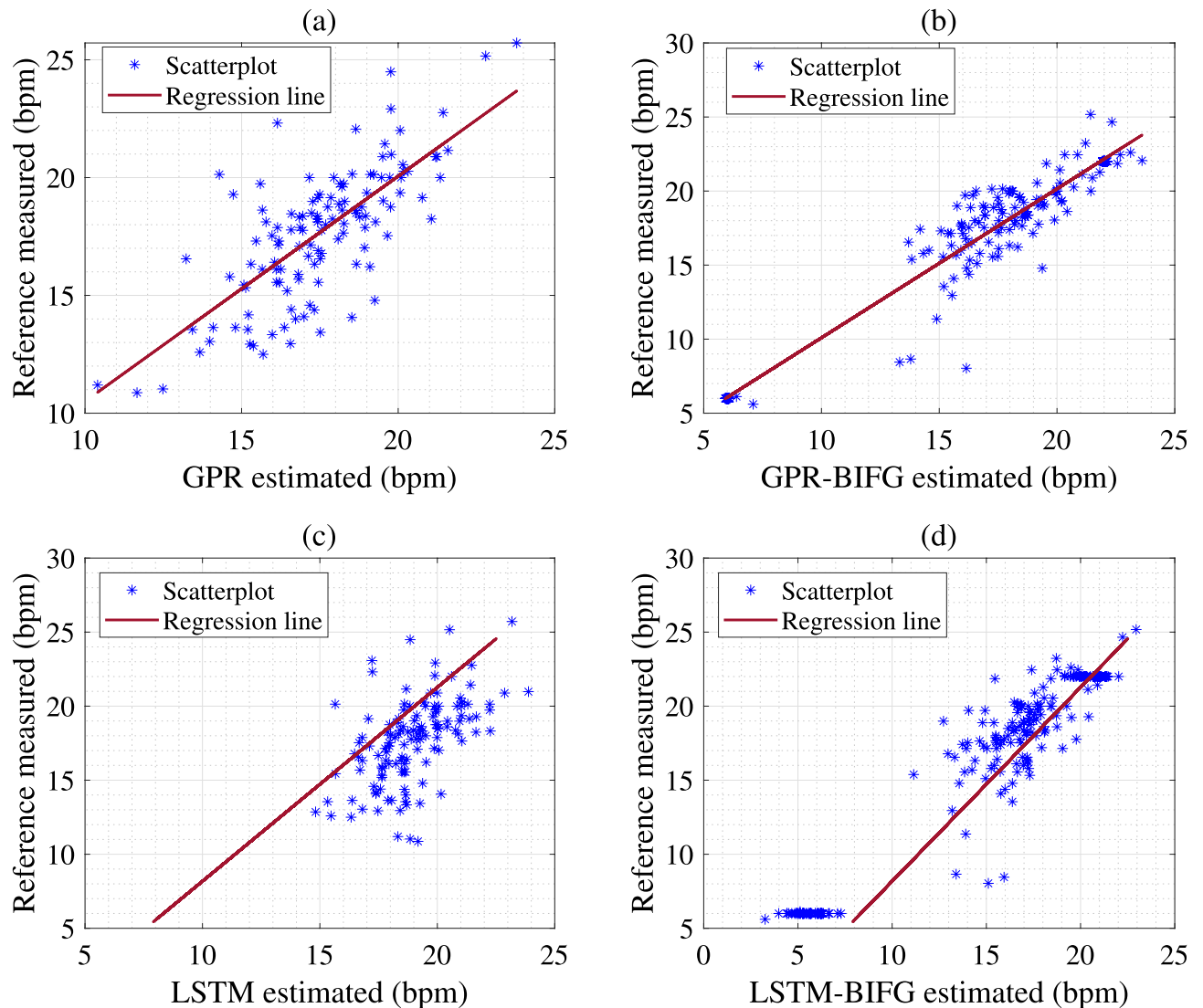


Fig. 8. Top panel (a) represents a R^2 results compared with the reference RR value acquired from the GPR; Panel (b) denotes a R^2 results compared with the reference RR value obtained from the GPR-BIFG; Panel (c) shows a R^2 results compared to the reference RR acquired from the LSTM; Panel (d) denotes a R^2 results compared to the reference RR acquired from the LSTM-BIFG based on the BIDMC data set. These results were acquired using the first experimental scenario.

Based on the RRSYNTH data set, the (SVM-BIFG:0.79) model obtained a stronger correlation with the response compared to the (SVM:0.38) model. The R^2 of the (NN:0.73) model indicated a similarly stronger correlation with the response variable when compared to the (NN-BIFG:0.75) model. The R^2 of the (GBA-BIFG:0.69) model indicated a higher correlation with the response variable than that of the (GBA:0.44) model. However, the R^2 of the (LSTM:0.52) model indicated low correlations with the response variables compared to the (LSTM-BIFG:0.72) model as presented in Table 5. The R^2 of the (GPR-BIFG: 0.93) model was the highest of all the models, indicating the strongest correlation with the response variable than that of the (GPR:0.90) model based on the RRSYNTH data set, as shown in Table 5.

Table 6 presented the CIs to indicate the uncertainties associated with the GPR-BIFG and GPR models for estimating RR based on the BIDMC and RRSYNTH data set. Figure 9 denoted the CIs to show the uncertainties associated with the GPR-BIFG and GPR models for estimating RR based on the BIDMC data set. The CIs was to show the uncertainties associated with the GPR-BIFG and GPR models for estimating RR based on the RRSYNTH data set as shown in Fig. 10.

Discussion

In this study, the combination of bootstrap-based imbalanced feature generation and GPR (GPR-BIFG) facilitated the estimation of RR and CI from PPG signals for the first time. The computational complexity of the proposed GPR-BIFG model was similar to that of the conventional model. We discovered that the LSTM model

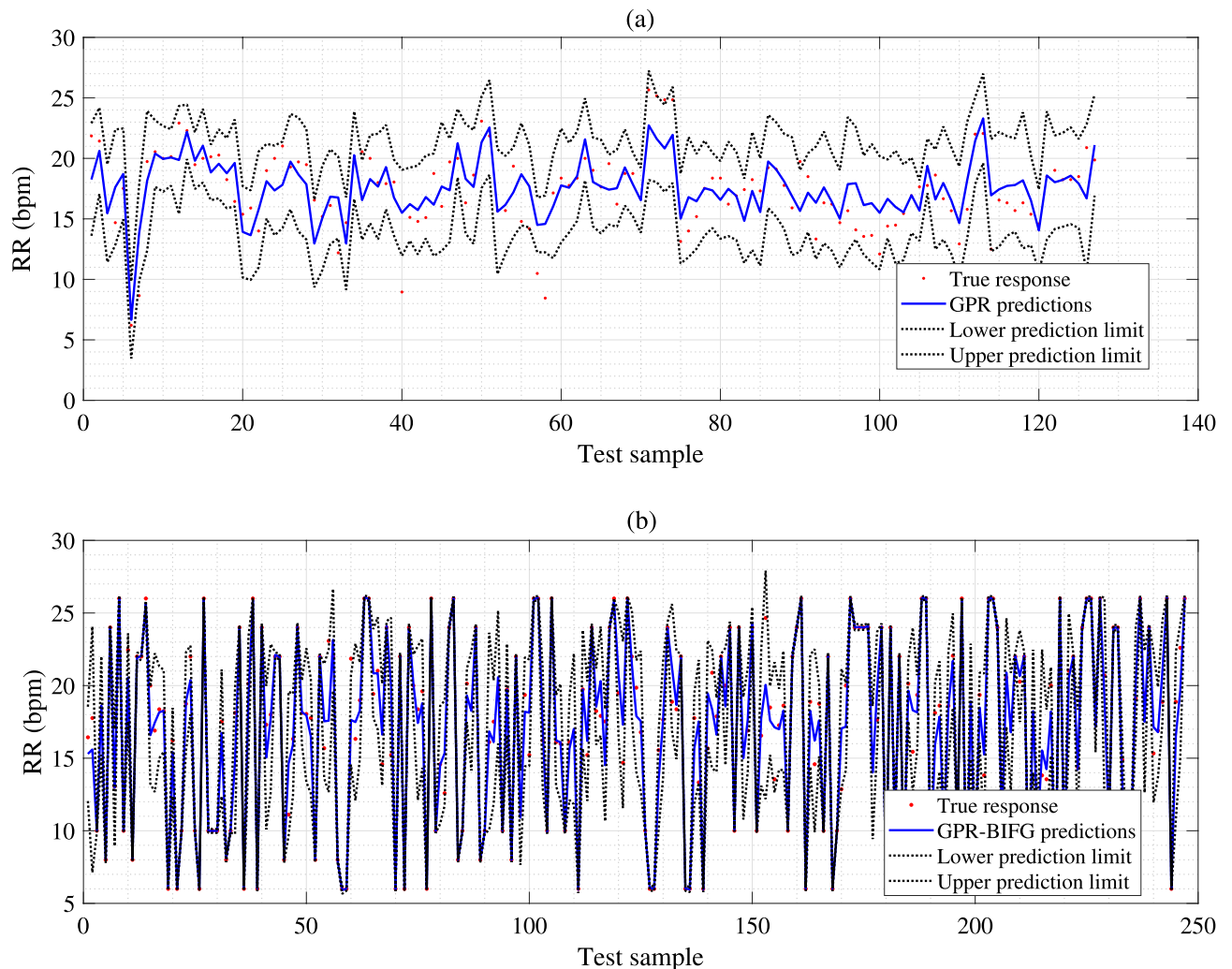


Fig. 9. Top panel (a) denotes the CIs estimation using the GPR model to represent the uncertainty for RR estimation; bottom panel (b) is the CIs estimation using the GPR-BIFG model to represent the uncertainty for RR estimation based on the BIDMC data set. These results were acquired using the first experimental scenario.

consumes a larger amount of computer resources than the traditional models, as shown in Table 2. Among all evaluated models, the proposed GPR-BIFG model had the best performance, with lower MAE than those of the SVM-BIFG, NN-BIFG, LSTM-BIFG, GBA-BIFG, and GPR algorithms. Furthermore, the GBA-BIFG model exhibited low MAEs and stable SDs. Overall, the proposed BIFG model yielded excellent results based on the BIDMC data set as shown in Table 3.

The proposed GPR-BIFG model, based on the RRSYNTH dataset, showed a slightly higher MAE (1.44 bpm) than the MAE (0.89 bpm) of GPR-BIFG using BIDMC dataset experiments. However, it outperformed the SVM-BIFG, NN-BIFG, LSTM-BIFG, GBA-BIFG, and GPR algorithms with lower MAE in the experiments using the RRSYNTH dataset as shown in Table 3. These results are believed to be caused by artificial data artifacts in the RRSYNTH dataset. In many samples of the RRSYNTH dataset, RRs were observed to be as high as 40–60 bpm. These results suggest that the proposed BIFG model is optimized to precisely balance data fit and smoothness. In addition, the proposed GPR-BIFG algorithm avoids overfitting, and the artificial power spectral curves generated using BIFG are fused to the tachypnea and hypopnea imbalance data, which means the data imbalance is resolved. Therefore, the proposed method helps accurately estimate RR, as shown in Fig. 7b and d.

The RMSE results also showed that the proposed BIFG model performed best when combined with the GPR learning algorithm. The RMSE results of GBA-BIFG, NN-BIFG, and LSTM-BIFG models were improved when combined with BIFG based on the BIDMC dataset, as shown in Table 4. According to the RMSE results in Table 4, the GPR-BIFG model (3.30 bpm) outperformed the SVM-BIFG (5.49 bpm), NN-BIFG (5.26 bpm), GBA-BIFG (6.55 bpm), and LSTM-BIFG (6.22 bpm) models based on the RRSYNTH dataset. Although the RMSE result of the SVM algorithm (8.40 bpm) was worse than other algorithms such as NN, GBA, LSTM, and GPR, when the SVM learning model was combined with the BIFG algorithm, the RMSE result (5.46 bpm) outperformed the GBA-BIFG (6.55 bpm) and LSTM-BIFG (6.22 bpm) algorithms because the number of imbalanced data significantly increased. Therefore, when the proposed BIFG model was combined with the conventional ML

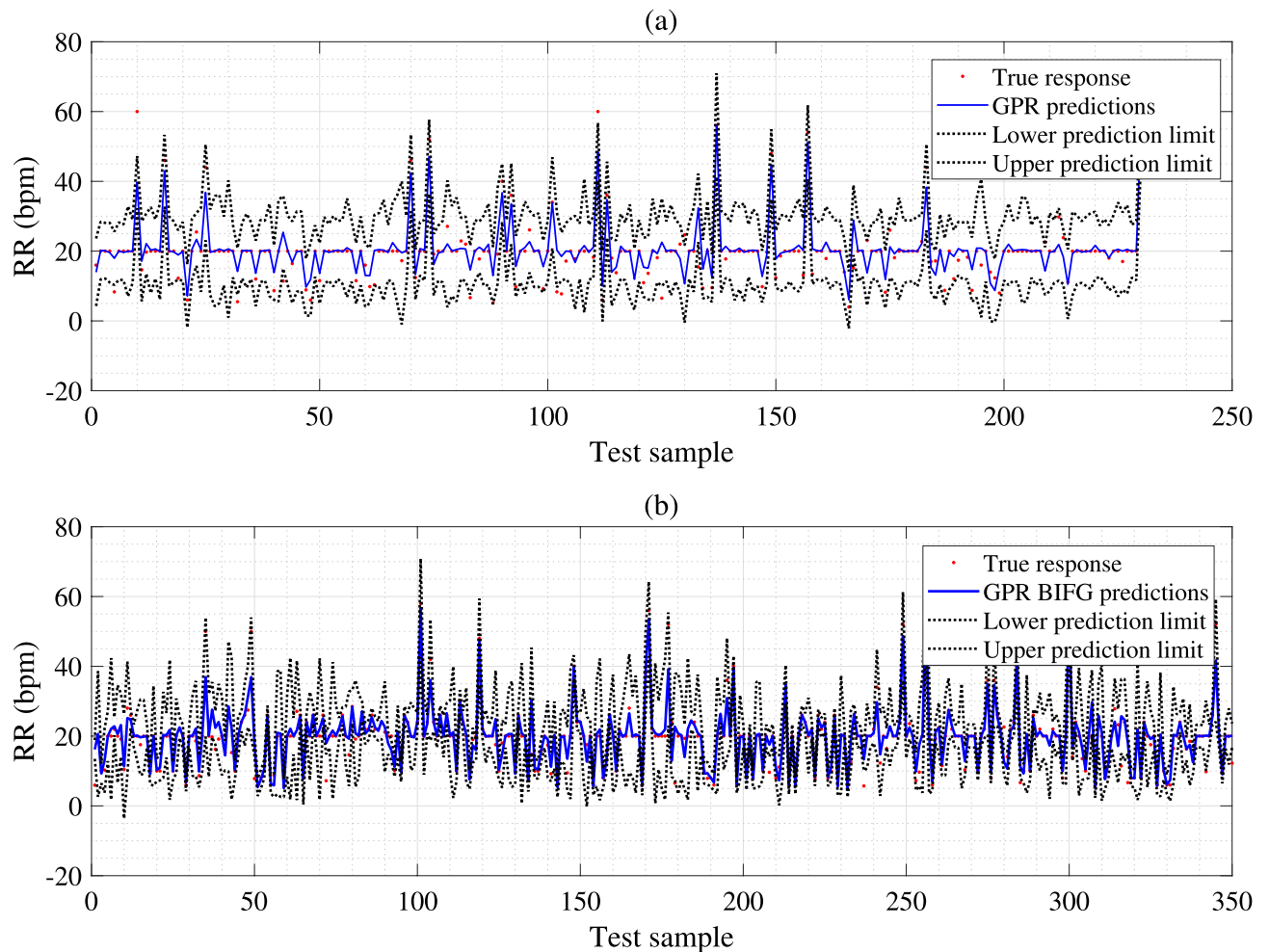


Fig. 10. Top panel (a) is the CIs estimation using the GPR model to represent the uncertainty for RR estimation; bottom panel (b) shows the CIs estimation using the GPR-BIFG model to represent the uncertainty for RR estimation based on the RRSYNTH data set. These results were acquired using the first experimental scenario.

model, we confirmed that it showed a lower RMSE than the conventional ML model without the BIFG model on the RRSYNTH dataset.

In addition, the R^2 values (0.94) and (0.95) of the SVM-BIFG and GBA-BIFG algorithms showed stronger relationships with the response than the R^2 values (0.63) and (0.59) of the SVM and GBA without the BIFG algorithm, as shown in Table 5. The R^2 (0.93) and (0.94) of the NN-BIFG and LSTM-BIFG algorithms showed higher relationships with the response variables than the R^2 (0.56) and (0.54) of the NN and LSTM algorithms, as shown in Table 5. The R^2 (0.96) of the proposed GPR-BIFG algorithm also showed higher relationships with the response variables than the existing GPR algorithm based on the BIDMC data set, as denoted in Table 5. We also confirmed that combining the BIFG algorithm with conventional learning models performed better in experiments using the RRSYNTH data set, as shown in Table 5. Therefore, we demonstrated that the proposed BIFG algorithm can improve the five learning algorithm's performance by addressing the imbalanced data problem.

In this study, we used the GPR and GPR-BIFG models to predict the CI for RR measurements, which allowed us to express the uncertainty of RR estimates for the first time. Table 6 represented that the GPR and GPR-BIFG models could obtain the CI using the model itself without combining other algorithms. However, conventional models require combining an algorithm to estimate the uncertainty to express the CI. The CI results of the proposed GPR and GPR-BIFG models were shown in Fig. 9, which can estimate the uncertainty of RR. The GPR model can estimate the uncertainty by providing the RR and CI estimates. However, in RR estimation, low bpm and high bpm respiration cannot be estimated due to the fundamental problem of imbalanced data, as shown in Fig. 7a.

On the other hand, the proposed GPR-BIFG method was suitable for estimating low bpm and high bpm respiration and estimating uncertainty by measuring CI based on the BIDMC dataset (see Fig. 9b). Therefore, the proposed method can overcome the problem of imbalanced data and improve performance. Typically, devices that measure RR only provide single-point estimates without a CI, making distinguishing between statistical and

physiological variations difficult. Consequently, the proposed BIFG can be applied to home-based monitoring systems for elderly patients. Statistically repetitive and widespread CIs in elderly patients can trigger alarms or alert nurses or primary care physicians in home-based monitoring settings⁴³. Therefore, predicting the CI for RR measurements can help improve reliability.

In the second RRSYNTH dataset experiment, as shown in Table 6, the CIs were estimated very wide, indicating that it is unsuitable for estimating RR fluctuations. These results imply that the estimation errors (MAE and RMSE) increase due to signal distortion and artifacts in the RRSYNTH dataset, and thus, the CIs also widen as shown in Fig. 10. Therefore, in future, we will continue to study how to reduce RR estimation errors through various data experiments and estimate CIs that can be applied to reliable home-based monitoring. We tested and evaluated the proposed BIFG algorithm's performance using two datasets. However, the clinical BIDMC dataset has limitations due to its relatively small number of participants and limited records. In addition, the RRSYNTH dataset is an artificial synthetic dataset with adequate recorded samples. However, we found that the RR was estimated to be more than 40 bpm due to severe data artificial artefacts.

The biggest challenge in developing a home-based monitoring device is the PPG signals for each patient. The PPG signal varies depending on the measurement location, skin texture, and motion artifacts. It also varies depending on the cardiovascular characteristics of elderly patients. In this paper, we implemented a bootstrap-based imbalance feature generation independent of these factors and showed equally good performance for all patients. The fusion power spectrum curves (GPR-BIFG: 0.96) showed that the model's coefficient of determination (R^2) was the highest among all models, indicating the strongest correlation with the response variable. This result improved the performance compared to the existing ML model without BIFG, as shown in Tables 3, 4, and 5.

Limitations and future work

Although we have proven our proposed method using the publicly available BIDMC and RRSYNTH datasets used in this study, we plan to validate our proposed BIFG method using a heterogeneous standard database MIMIC-III (12,000 records) with diverse cases in various clinical conditions in future studies. We will validate our proposed methodology on such an extensive database to provide reliability for clinical use. In this way, we will enhance life monitoring systems' reliability to mobile and home-based healthcare devices.

The uncertainty of RR for reliability improvement is expressed through CI. However, the CI obtained from the RRSYNTH data set is calculated based on the distribution of RR estimates, so it showed a relatively wide CI. This is because the RR estimate error increased compared to when BIDMC was used. Hence, in future studies, we will continue to research how to reduce the robust RR estimate error in the RRSYNTH data set, which includes various artificial distortions. A relatively wide CI increases the probability that RR estimates are included, but it has the disadvantage of lowering the reliability of the RR monitoring system. In addition, if the CI is too narrow, even a minimal change in the RR estimate can easily deviate from the CI, which may limit its role as an RR monitoring system. Therefore, in future studies, we will secure an appropriate range of CI to overcome these shortcomings. We used only single PPG signals to reduce the hardware complexity of continuously monitoring biosignals, power efficiency should also be considered for mobile applications. Currently, machine and deep learning approaches are computationally very intensive. Therefore, we will develop a simple machine-learning model to improve the algorithm complexity and reduce the device.

Although we demonstrated strong results using public datasets, our model should be tested on real-time, patient-specific data to further validate its robustness. While our approach is more efficient than deep learning models, real-time deployment in edge devices requires further optimization. Future work could explore hybrid models combining BIFG-GPR with deep learning, or domain adaptation techniques to improve generalization across different datasets.

Conclusion

This study proposes a novel method to estimate RR and CI using the GPR-BIFG model based on PPG signals. The input PPG signal was segmented, and continuous features were extracted using the autocorrelation function-based power spectral feature extraction technique. The imbalanced distribution features of the PPG signal, which were the focus of this study, were supplemented by generating artificial feature curves using the bootstrap technique. We trained the proposed GPR-BIFG model to overcome the overfitting and data sample imbalance problems by combining the original feature curves and the artificially generated feature curves. This approach achieves high stability and accuracy by randomly mixing the original feature curves and the artificial feature curves. The proposed GPR-BIFG algorithm measures the CI, which represents the level of uncertainty that may occur due to the physiological variability of the PPG signal. Therefore, the proposed GPR algorithm is a reliable model for estimating RR and CI accurately. We will conduct future studies by expanding the dataset, using cross-validation with a larger cohort, and applying the proposed method to heart rate and blood pressure variability. We will validate the proposed technique using another public dataset (MIMIC-III). Thus, the proposed GPR-BIFG model can be used to improve the performance of clinical home-based monitoring systems and design a reliable framework. These simple wearable smart devices can estimate RR and CI and are convenient to wear and carry for long periods.

Data availability

The datasets used and analysed during the current study are available from the corresponding author on reasonable request.

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Author contributions

Soojeong Lee: Conceptualization, Methodology, Software, Formal analysis, Investigation, Resources, Writing - original draft, Visualization, Writing—review & editing, supervision. Gyanendra Prasad Joshi: Methodology, Investigation, Resources. Gangseong Lee: Formal analysis, Resources, Writing—review & editing.

Declarations

Conflict of interest

The authors declare no competing interests.

Additional information

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